NRSC 7657 Workshop in Advanced Programming for Neuroscientists
Instructors: Daniel Denman, John Thompson

Course Overview

The goal of this in-person course is to provide neuroscience students with practical skills in modern programming to facilitate their research and understand technical development in the field. It is project-based mixed with a flipped classroom for practical pedagogy in computational tools and techniques for a broad range of topics within neuroscience. The course will cover concepts in programming, data types, workflow, data and code management, and collaboration tools. Students are expected to have completed some basic introduction to programming prior to this course, or have previous experience with scientific computing in either MATLAB or Python. Students are expected to complete an independent project using Python or (/and) MATLAB, using data from their own work or publicly available datasets. The course will use both Python and MATLAB for didactic sessions, with the goal of providing basic familiarity in both widely used platforms in neuroscience. The course is taught over 10 weeks, with one 2 hour session per week. Each session is a "flipped classroom", in which students are expected to prepare and drive session content with questions. All students will create and maintain a GitHub repository for their final project.

Schedule

June 9- Aug 22 2025
Monday 10AM –12PM, RC1S 8108

June 9	Week 1 – overview / review	Course overview: theory of computing, landscape of computing options. Basic usage in python and MATLAB; basic data types; environments Style guidelines (ten simple rules); git and version control
June 16	Week 2 – language fundamentals	Functions; Objects and Classes; Workspaces Typical data formats: working with tabular data, images, and time series. NeurodataWithoutBorders format
June 23	Week 3 – workflow management and outputs	Importing and exporting Packaging
June 30	Week 4 – usability	Plotting and visualization - from bar charts to 3D animation
July 7	Week 5 – scaling	Troubleshooting and debugging; unit testing
July 14	Week 6 – collaboration	Iteration and code profiling; parallel computing. Code quality-of-life topics
July 21	Week 7 – applications/flex topic	Cloud-based tools: AWS, GCC, Colab, jupyterhub, deepnote. Overview of some available SAAS tools, python focused.
		Group programming time
July 28	Week 8 – Open topics	Make-up session – open topics
August 4	Week 9 – Final presentations and code review	Final pres. and code review
		Final pres. and code review
August 11	Week 10 – Final presentations and code review	Final pres. and code review
		Final pres. and code review

Evaluation

Students will be evaluated on class attendance, participation, and effort towards their final project. *Weekly* progress reports on the final coding project are expected to be uploaded to the project GitHub repository.

Land Acknowledgement

As we gather, we honor and acknowledge that the University of Colorado's four campuses are on the traditional territories and ancestral homelands of the Cheyenne, Arapaho, Ute, Apache, Comanche, Kiowa, Lakota, Pueblo, and Shoshone Nations...The University of Colorado pledges to provide educational opportunities for Native students, faculty and staff and advance our mission to understand the history and contemporary lives of Native peoples.

Labor Acknowledgement

As we gather, it is important to acknowledge that much of what has been built in this country, including culture, economic structures, and physical structures has been made possible by the labor of enslaved Africans, including their ascendants and descendants, who suffered the horror of the transatlantic trafficking, chattel slavery, and Jim Crow. We acknowledge we are indebted to their labor and their sacrifice and that we are still experiencing the reverberations of this violence today through continued acts of violence, systemic racism, and sanctioned exploitation. We also acknowledge other acts of enslavement specific to Colorado inclusive of enslavement of Mexican persons in the San Luis Valley, enslavement of Mexican and Asian railroad workers, and the internment of Japanese persons during World War II.